Irrational Numbers

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Rational and irrational numbers

What is a rational number?

- A **rational** number is a number that can be written as a fraction in its simplest form
  - It must be possible to write in the form $\frac{a}{b}$, where a and b are both **integers**
    - b cannot be zero
  - This includes all terminating and recurring decimals
    - E.g. 0.15 (which is $\frac{15}{100}$) and 0.151515151515... (which is $\frac{15}{99}$)
  - This also includes all integers
    - 5 can be written as $\frac{5}{1}$
    - -3 can be written as $\frac{-3}{1}$ or $\frac{3}{-1}$
    - 0 can be written as $\frac{0}{1}$ (it is ok to have 0 in the numerator)

What is an irrational number?

- An **irrational** number is a number that **cannot** be written in the form $\frac{a}{b}$, where a and b are **integers** and $\frac{a}{b}$ is in its simplest form
  - A decimal which is non-terminating and non-recurring is an irrational number
  - The number $\sqrt{n}$, where n is not a square number, is an irrational number
    - This is also known as a **surd**

What irrational numbers should I know?

- You may be asked to identify an irrational number from a list
- Irrational numbers that you should recognise are π, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$,

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- For example 2π, $5\sqrt{2}$, $\frac{1}{4}\sqrt{5}$ are all irrational
  - Be careful: $\sqrt{2} \times \sqrt{2}$ is not irrational as it equals 2
- Most calculators will show irrational numbers in their exact form rather than as a decimal

Examiner Tips and Tricks

If you're not sure if a number is rational or irrational, type it into your calculator and see if it can be displayed as a fraction.

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